

Ethanol Blending in Gasoline - Panama

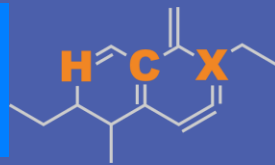
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**U.S. GRAINS &
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline - Panama



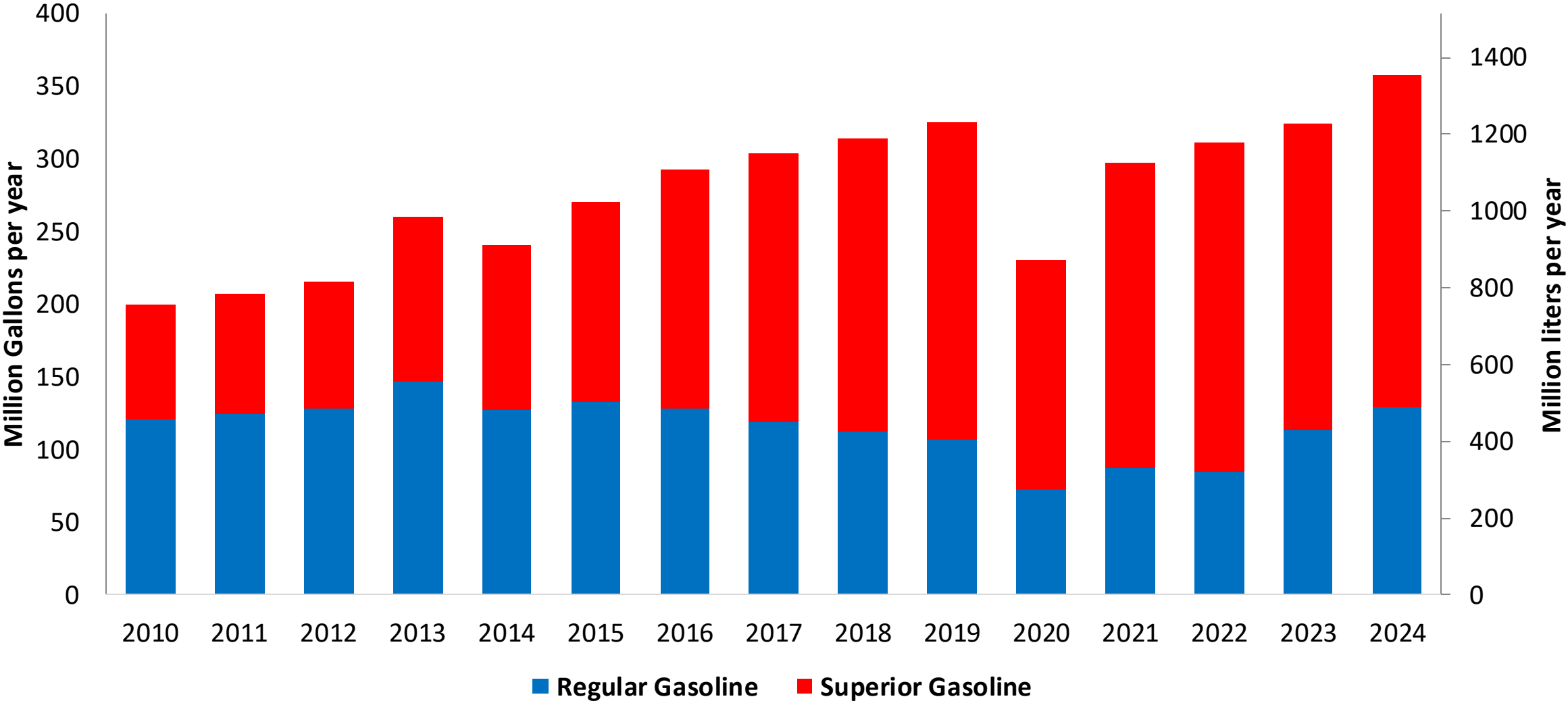
En 2024 demand was 358 million gallons (1356 million liters). Regular gasoline (RON 91) represented 36% of total demand and Superior gasoline (RON 95) 64%. Gasoline is supplied only with imports, mainly from United States, the Netherlands and Germany.

In 2011, 2% v/v ethanol blends with gasoline were authorized, with a planned gradual increase until reaching 10% v/v by 2016. However, in 2013 it was limited to 5% v/v and by 2014 blending stopped because the only national producer announced its production cease due to political dispute. A new mandate implementation has been approved for E10 and is expected to be implemented between 2026 and 2027.

Source: Secretaría de Energía - Panama

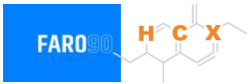
Gasoline Demand in Panama

Gasoline Demand by Grade in Panama

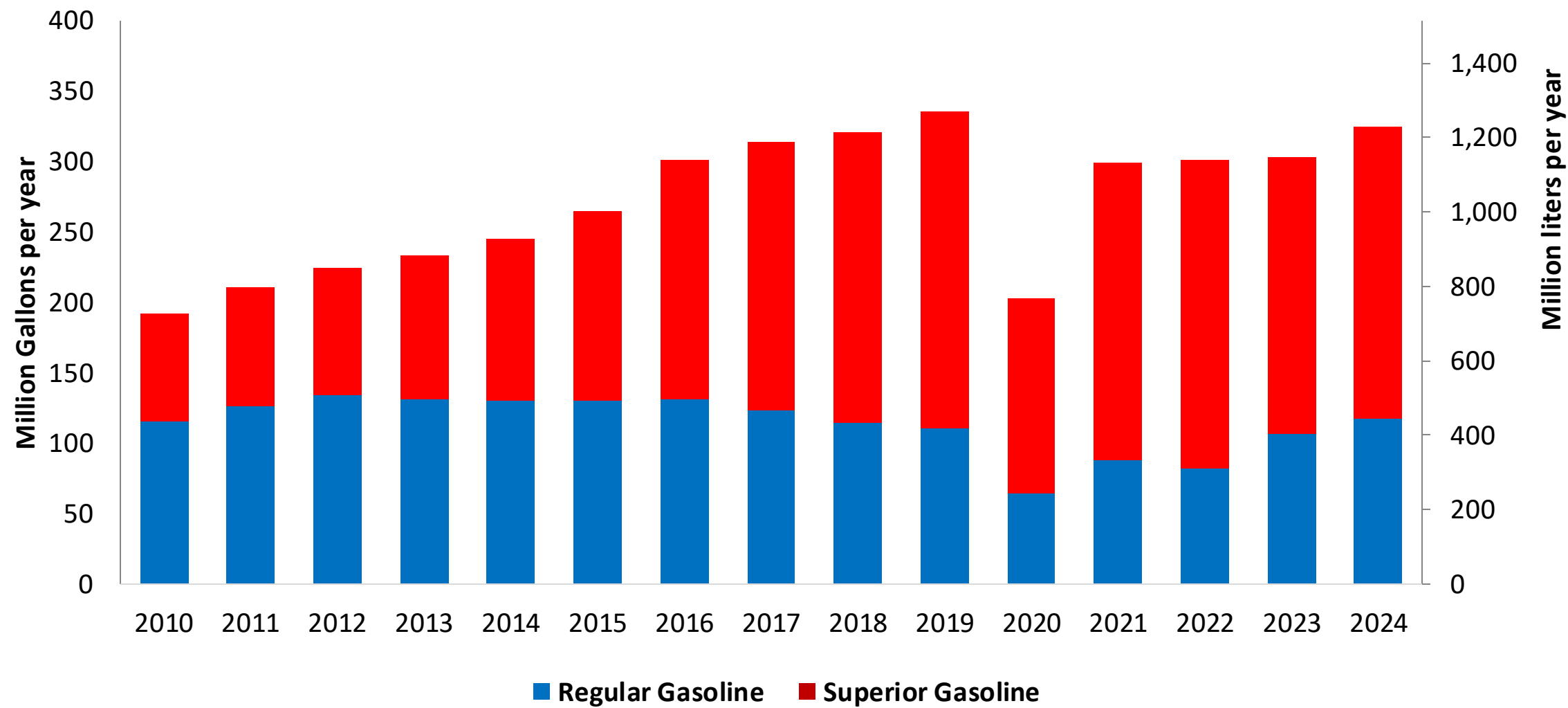


Source: Secretaría de Energía - Panama

Gasoline Imports to Supply Demand in Panama

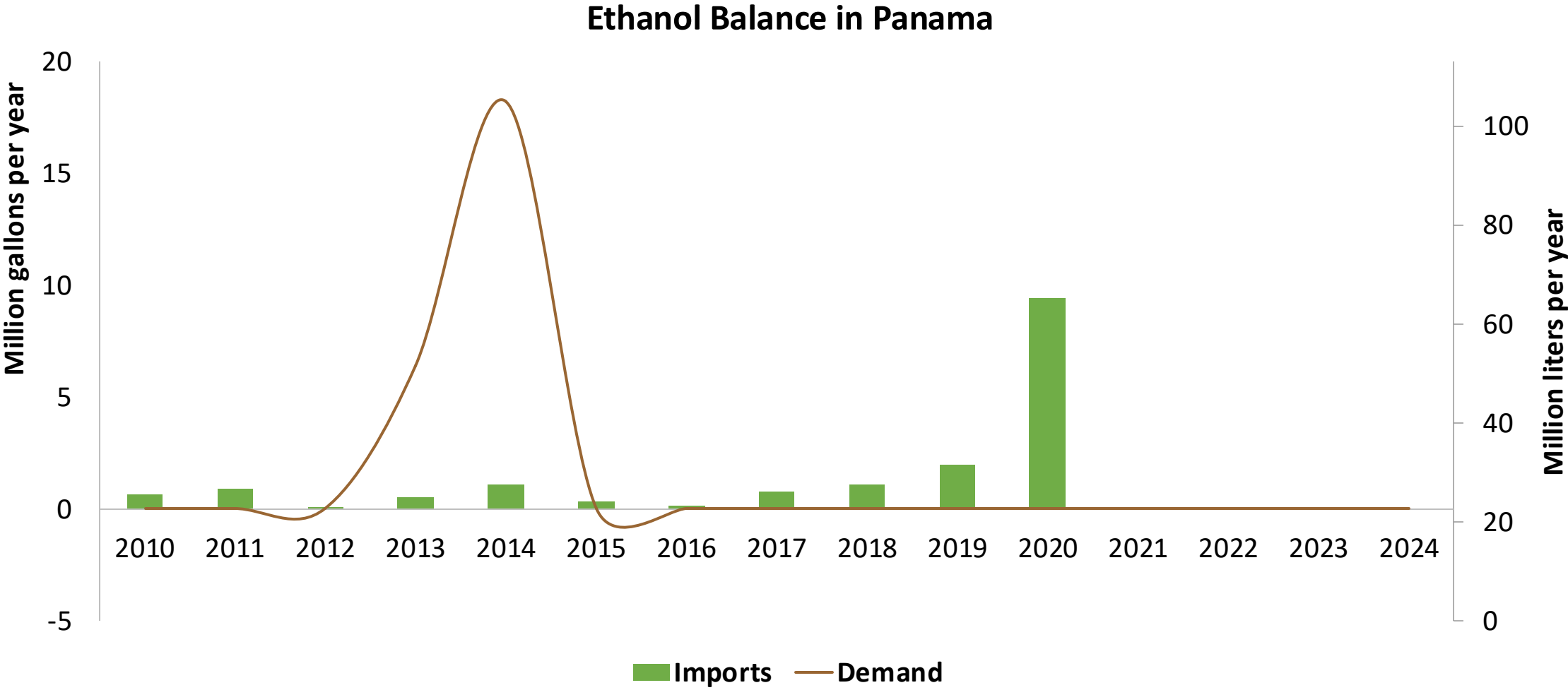


Gasoline Imports by Grade to Panama



Source: Secretaría de Energía - Panama

Ethanol Balance in Panama



Source: Secretaría de Energía - Panama

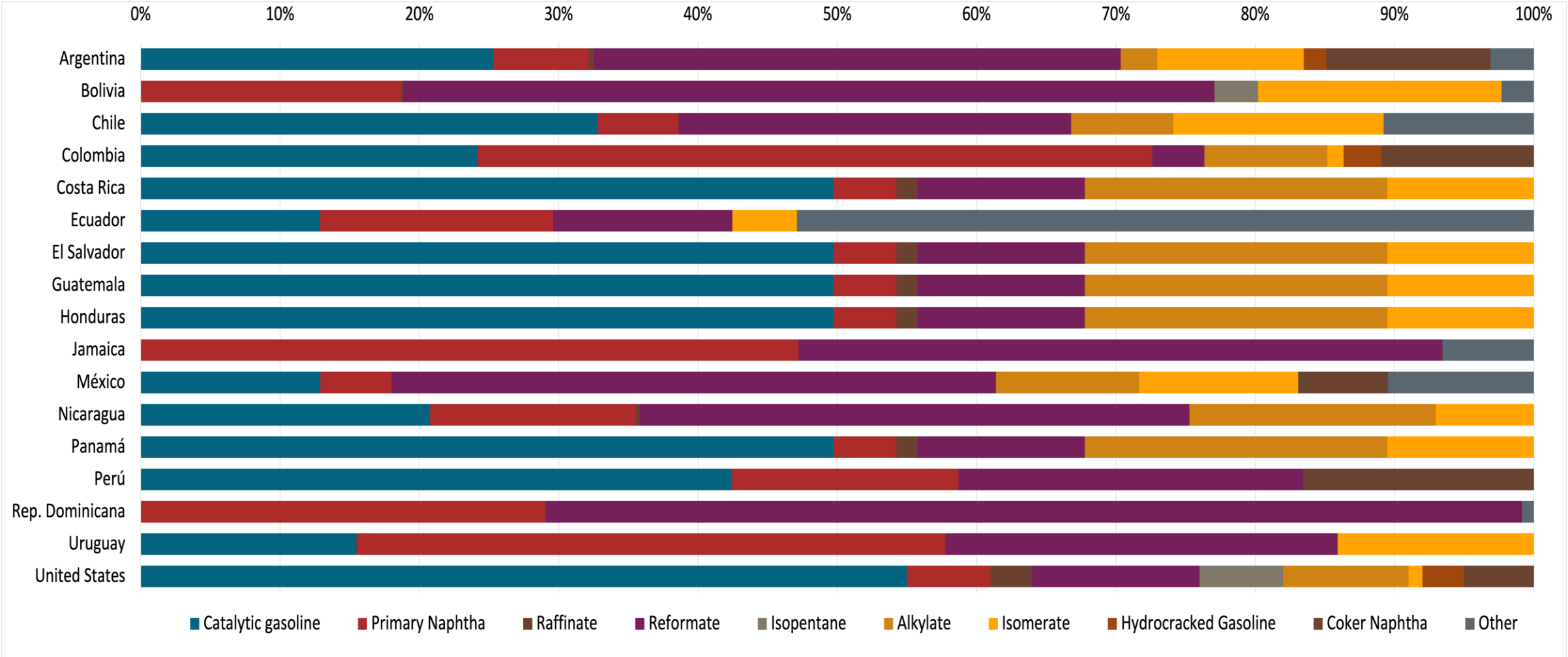
Gasoline Quality in Panama

Name	DGNTI-COPANIT 83-2013/ Resolution 425/2020		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2014/2021		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	RON 91	RON 95	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 5% v/v / <1,5% v/v	< 5% v/v / <1,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 50% v/v	< 50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 30% v/v	< 30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	-	-	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg / < 150 mg/kg	< 500 mg/kg / < 150 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	-/ < 0,7% v/v	-/ < 0,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	10% v/v	10% v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa (< 76 kPa if ethanol is added)	< 69 kPa (< 76 kPa if ethanol is added)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	0% v/v if ethanol is added	0% v/v if ethanol is added	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: RTCA

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



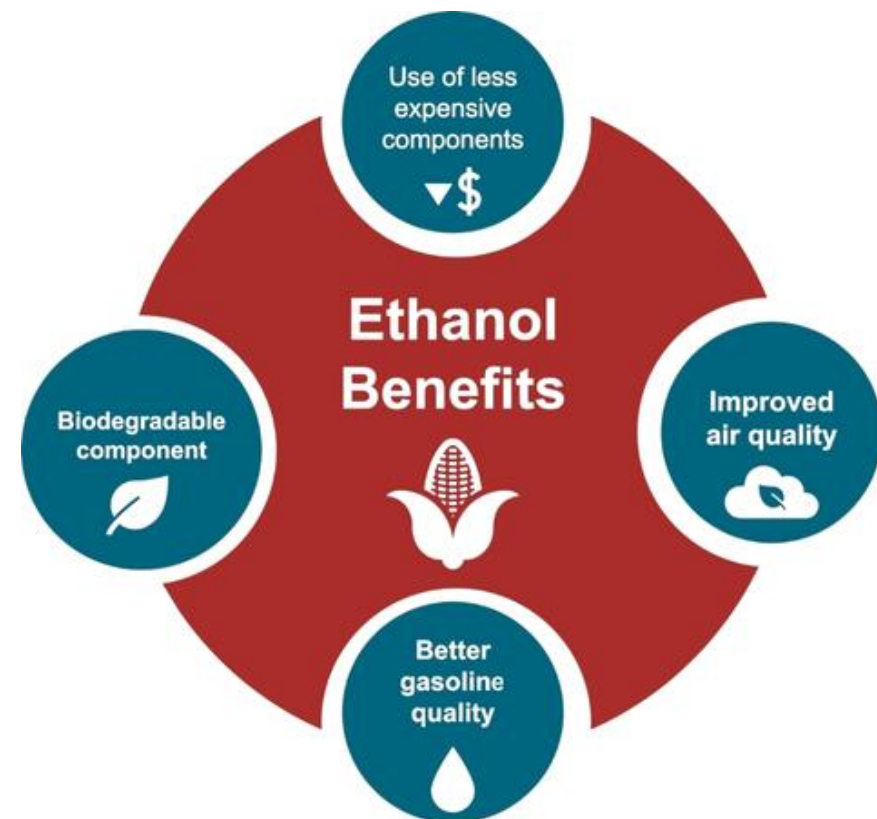
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

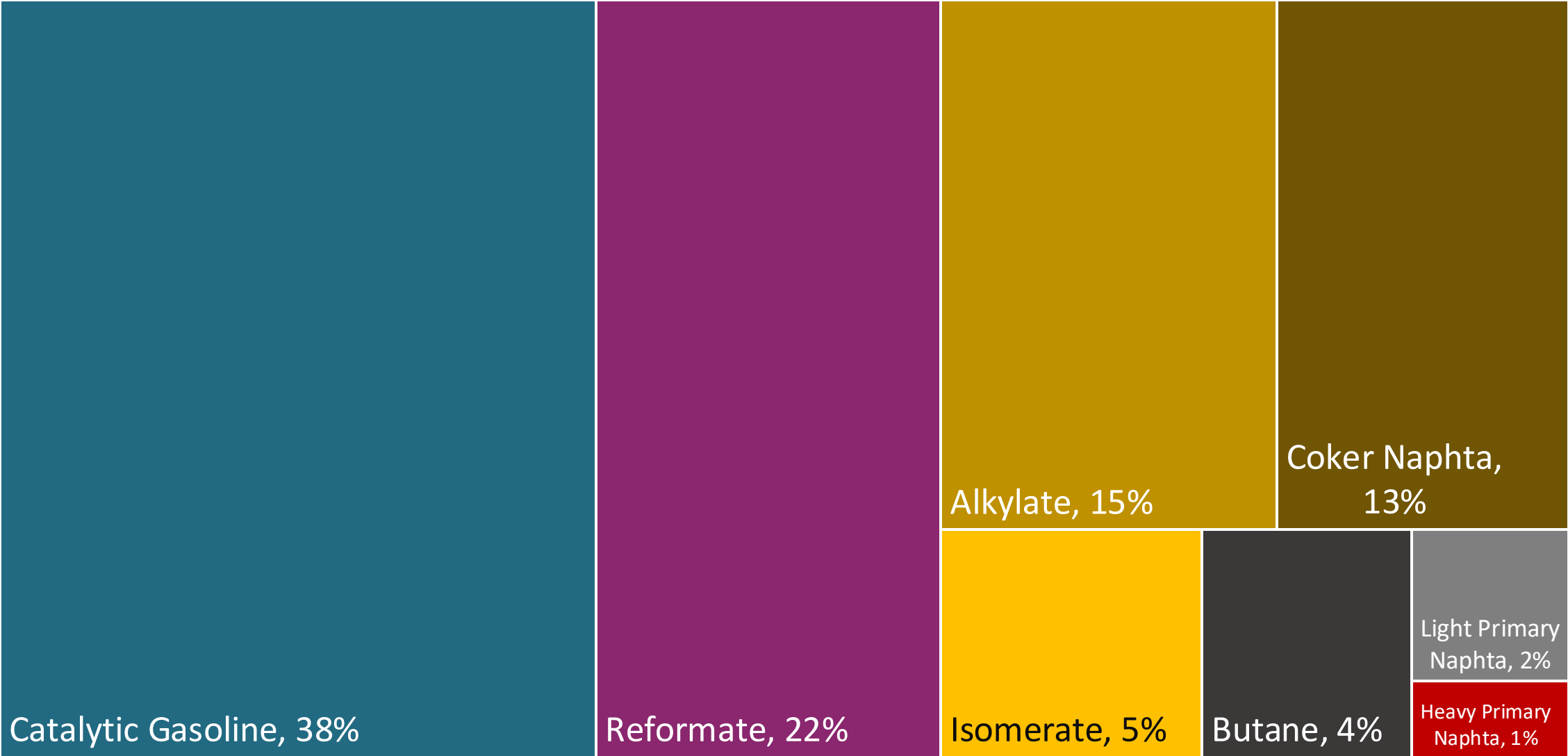
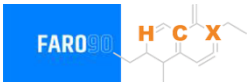
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

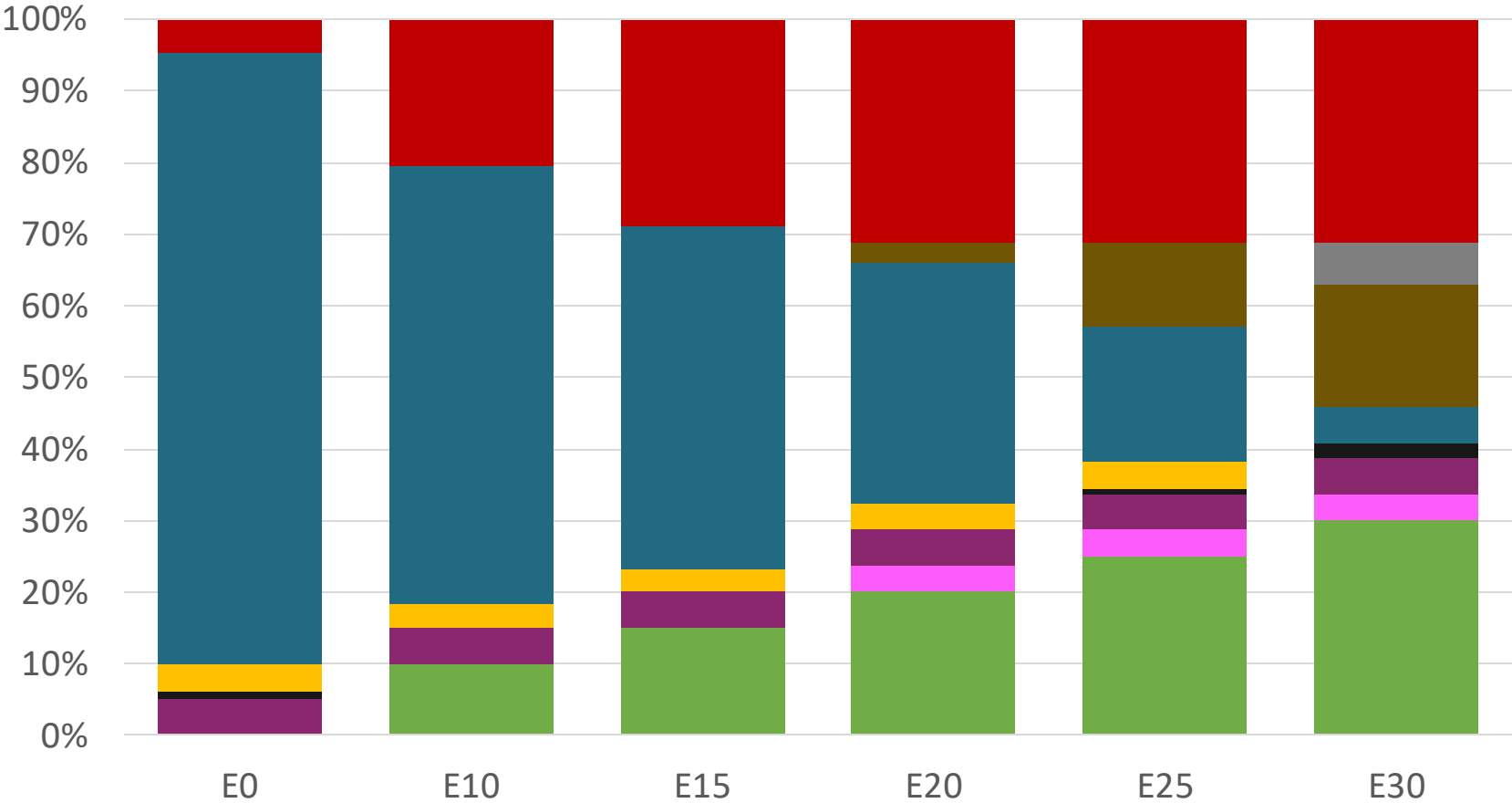
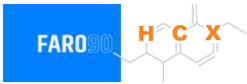


Blending Components - Panama



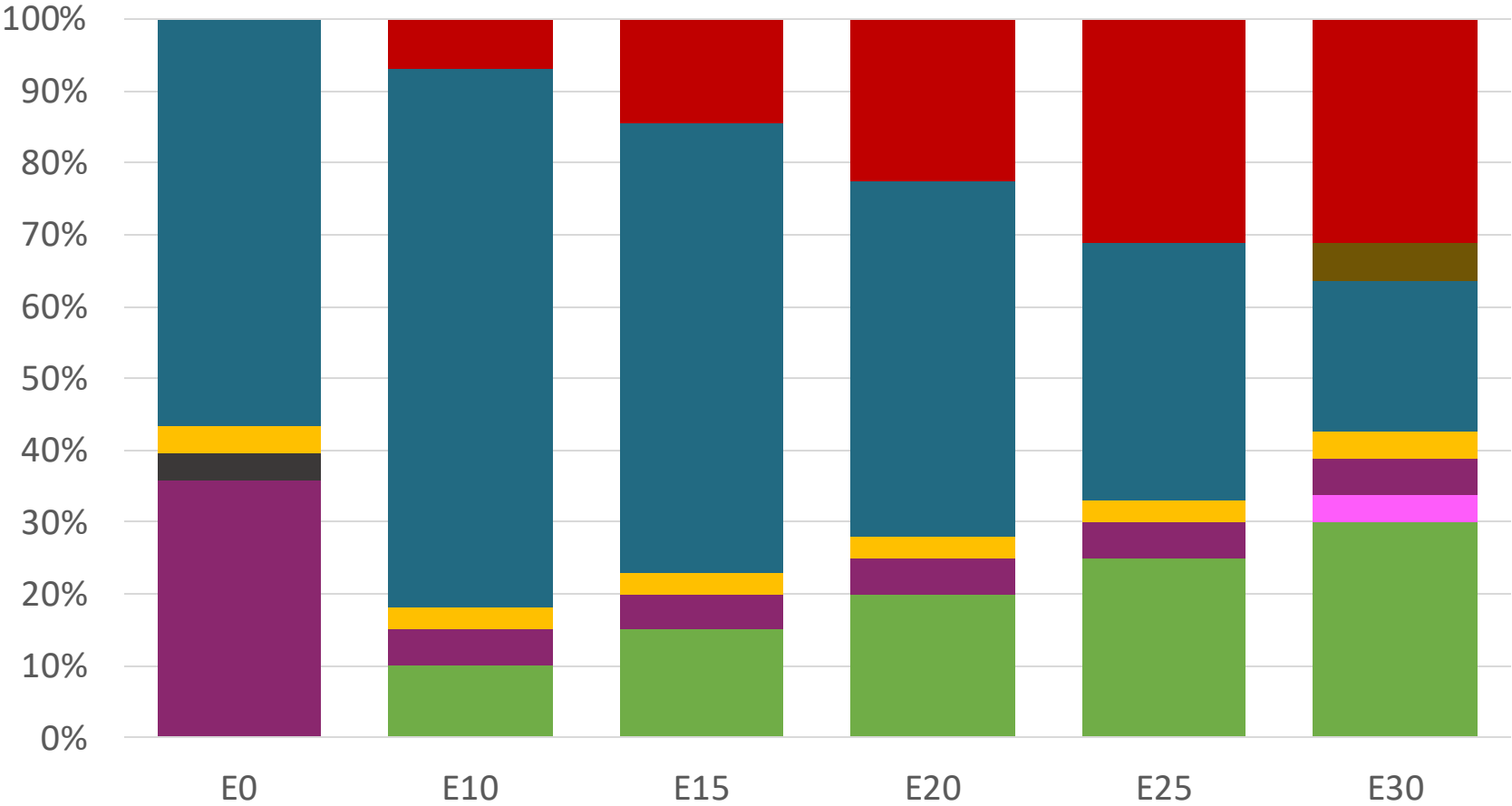
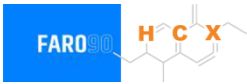
Source: Faro90

Panama – Regular – Constant Octane



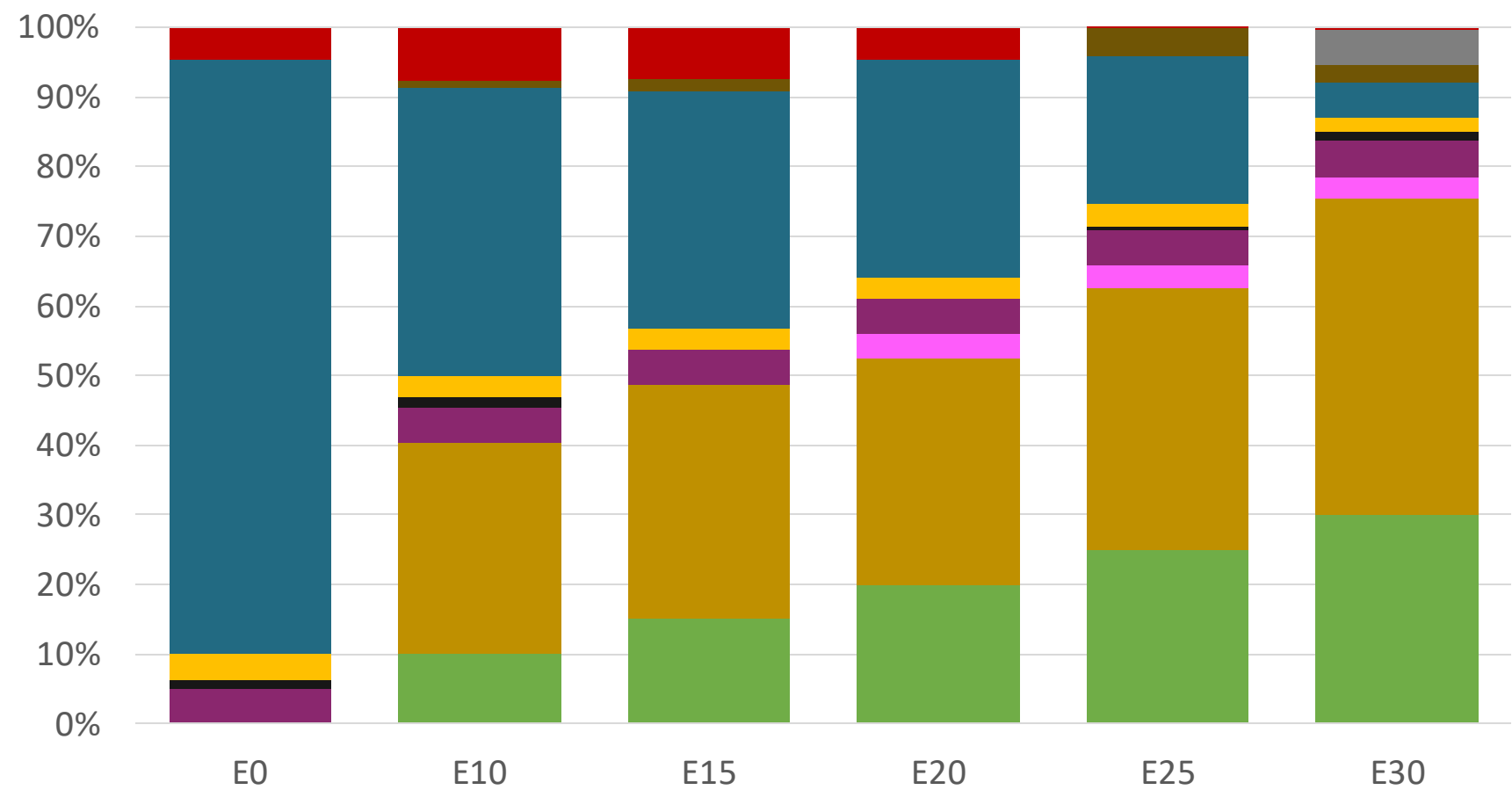
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.286	\$ 2.028	\$ 1.884	\$ 1.737	\$ 1.586	\$ 1.468

Panama – Premium – Constant Octane



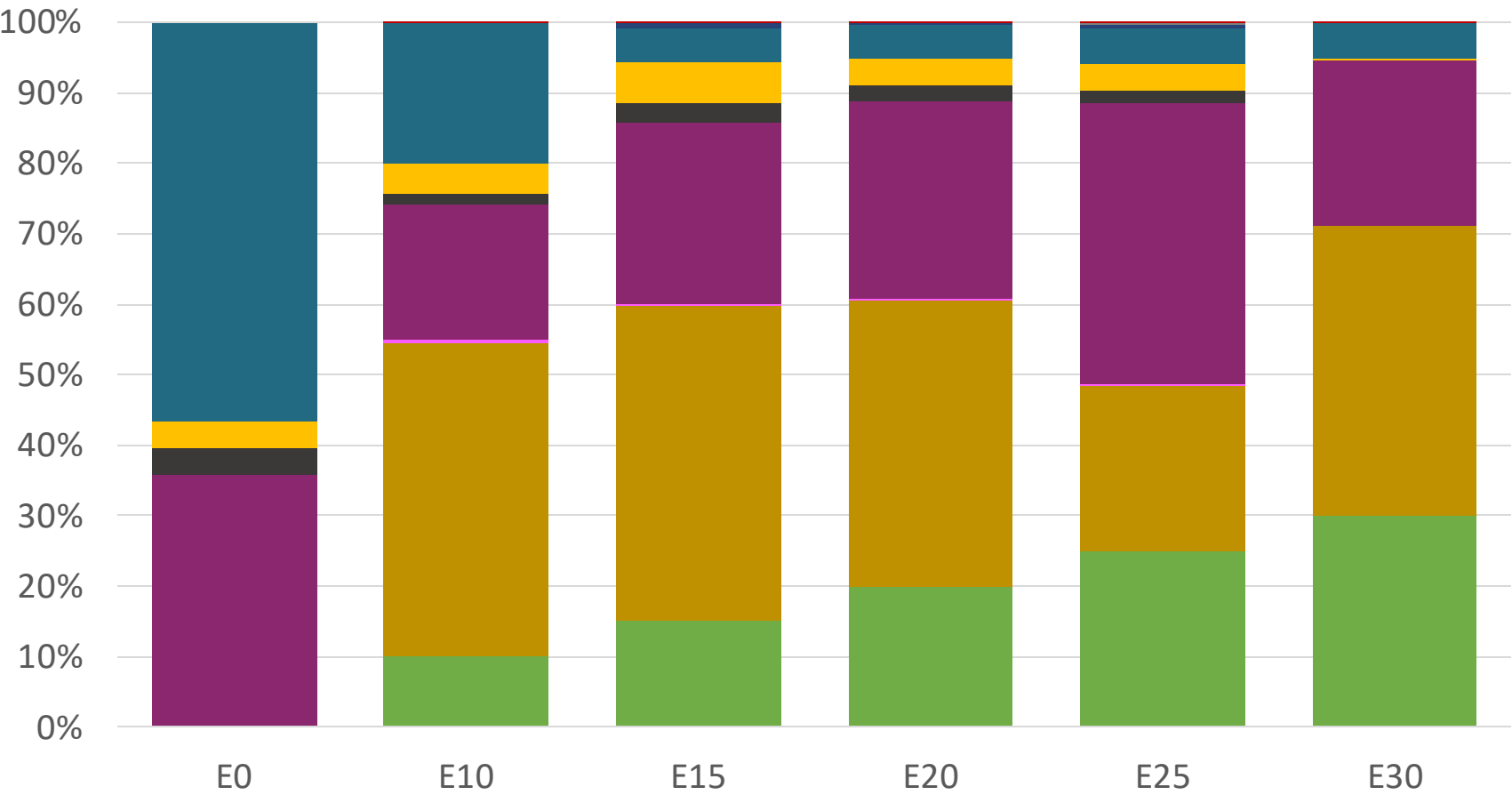
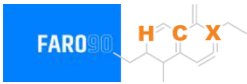
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.536	\$ 2.233	\$ 2.099	\$ 1.957	\$ 1.807	\$ 1.659

Panama – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.286	\$ 2.286	\$ 2.286	\$ 2.286	\$ 2.286	\$ 2.286

Panama – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

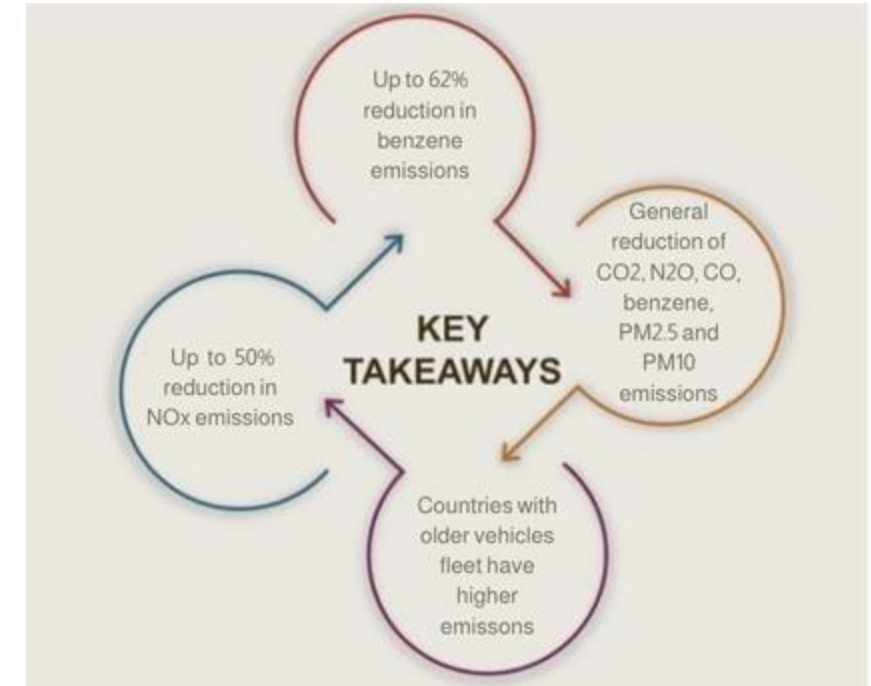
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

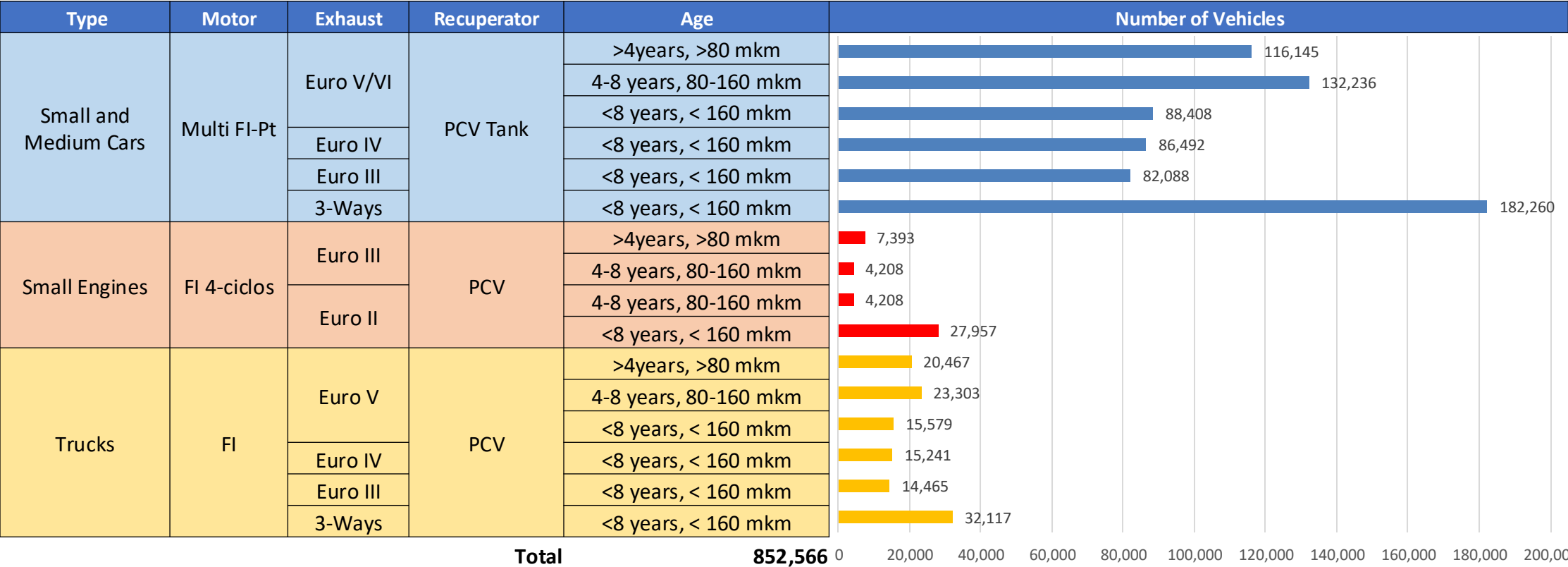
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicular Fleet in Panama

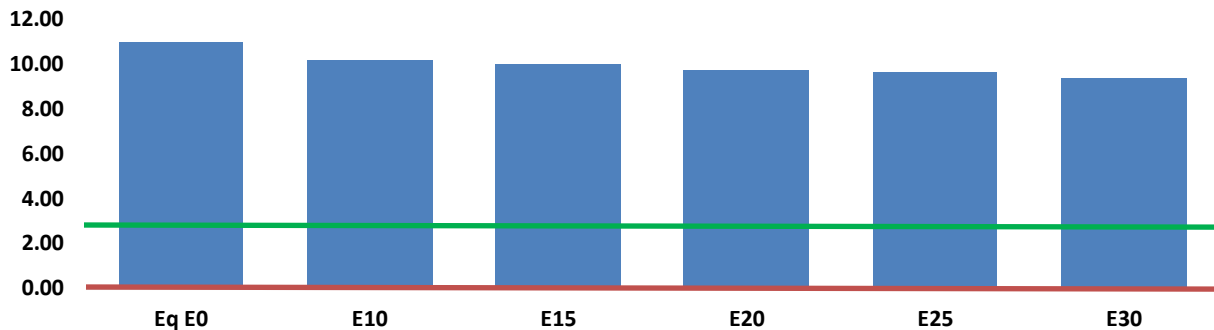


Vehicle Fleet: **852,566**
Average Age: **11.5 years**
Motorcycles: **5%**

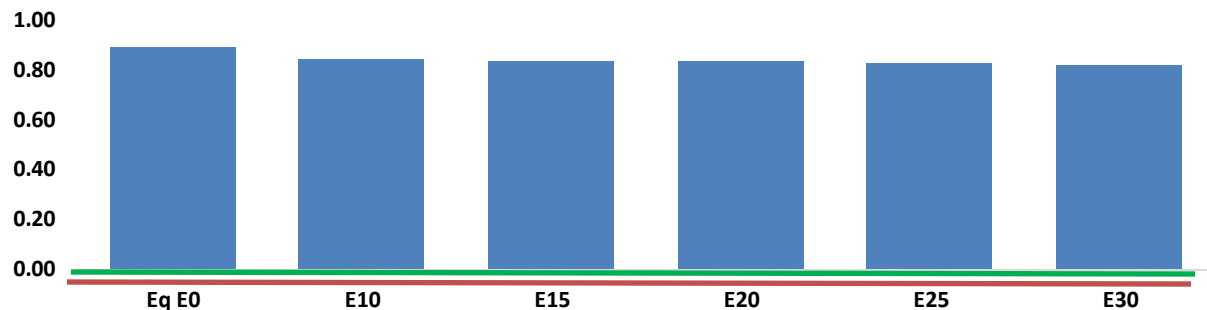
The FARO90 logo is shown in a blue box, and a chemical structure is displayed next to it.

United States
Euro 6 Standard

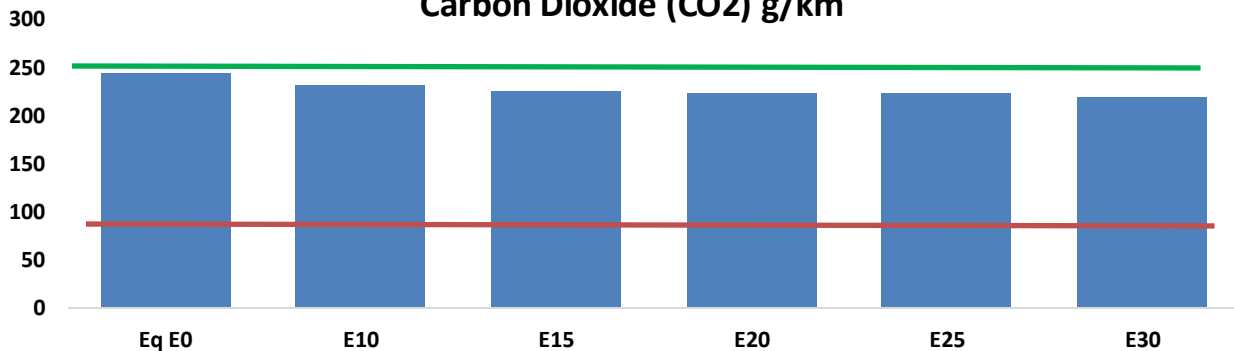
Carbon Monoxide (CO) g/km



Volatile Organic Compounds (VOC) g/km



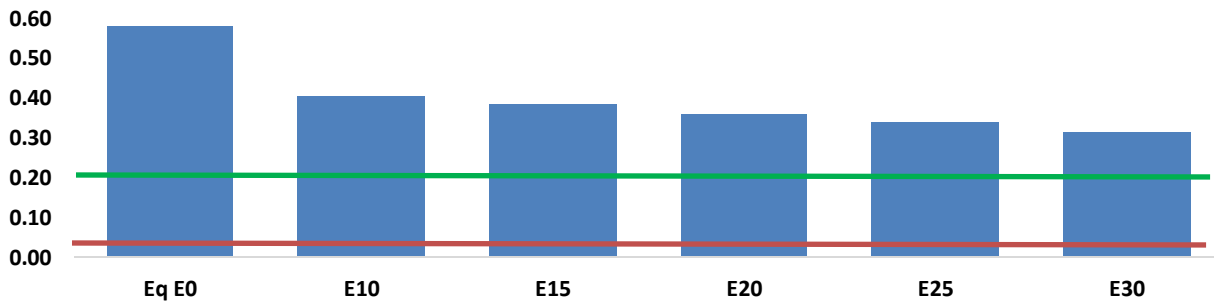
Carbon Dioxide (CO2) g/km



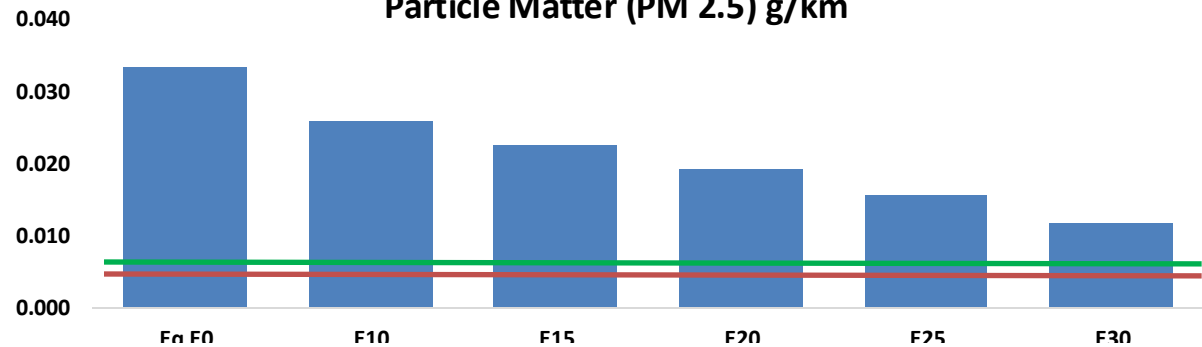
Aromatics (BTX) g/km



Nitrogen Oxides (NOx) g/km



Particle Matter (PM 2.5) g/km



Panama – Vehicle Emissions

Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	240,461	231,786	223,111	217,916	213,569	210,493	205,776	-7%	-11%
CO2	5,305,431	5,169,006	5,040,160	4,938,826	4,888,682	4,867,636	4,777,912	-5%	-8%
VOC	19,390	18,921	18,452	18,266	18,171	18,126	17,968	-5%	-6%
NOx	12,640	9,480	8,848	8,339	7,864	7,341	6,787	-30%	-38%
SOx	68	63	58	53	49	45	40	-15%	-28%
NH3	1,505	1,490	1,475	1,482	1,498	1,510	1,520	-2%	0%
N2O	190	189	188	189	193	195	197	-1%	2%
CH4	4,279	4,279	4,279	4,343	4,437	4,519	4,596	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	119	116	113	112	112	112	111	-5%	-6%
Acetaldehyde	229	263	385	587	799	913	1,079	68%	249%
Formaldehyde	756	735	856	1,005	1,053	1,165	1,269	13%	39%
Benzene	364	350	332	327	324	310	300	-9%	-11%
PM2.5	398	353	309	268	228	185	140	-22%	-43%
PM10	330	293	256	223	189	153	116	-22%	-43%

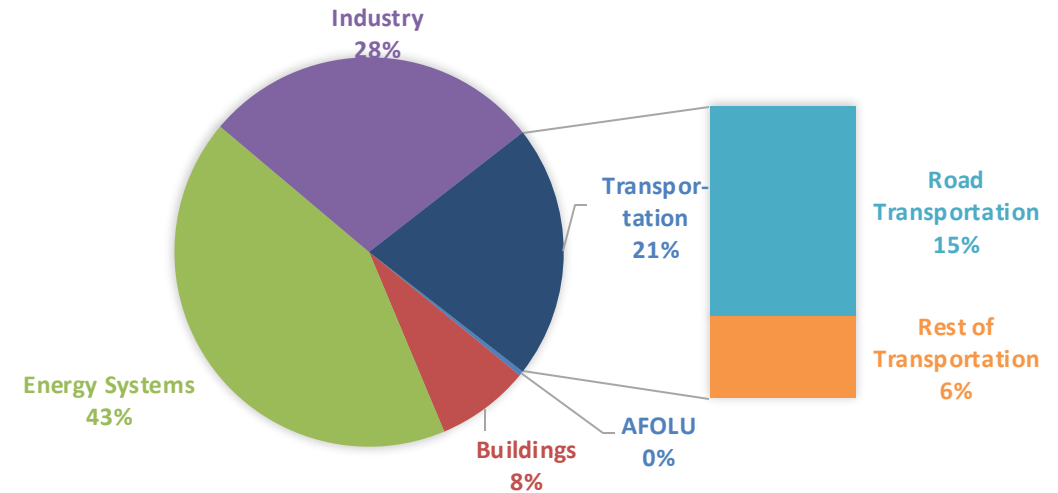
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

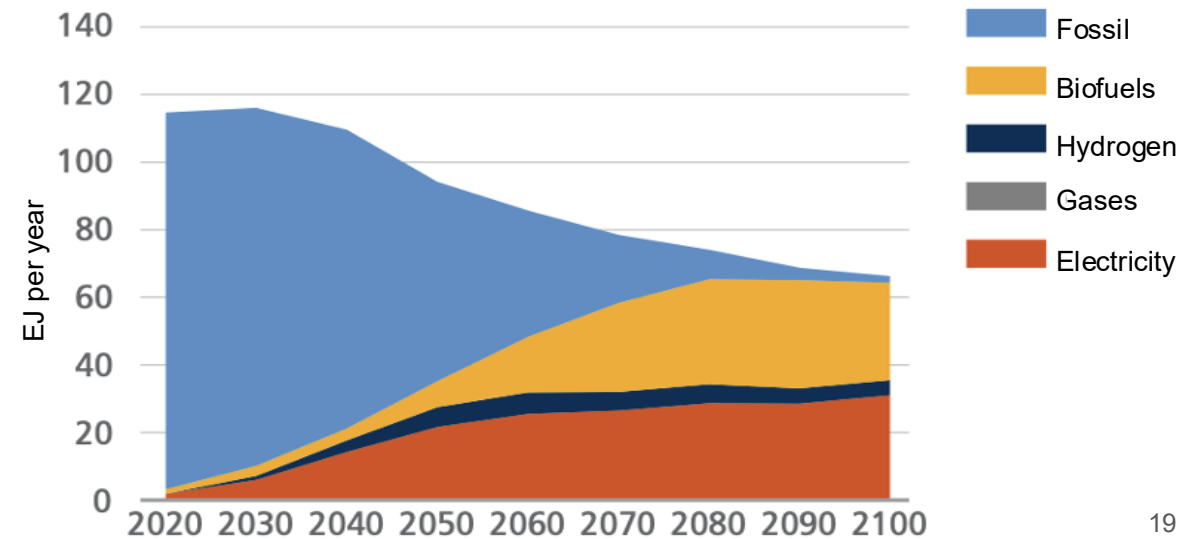
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

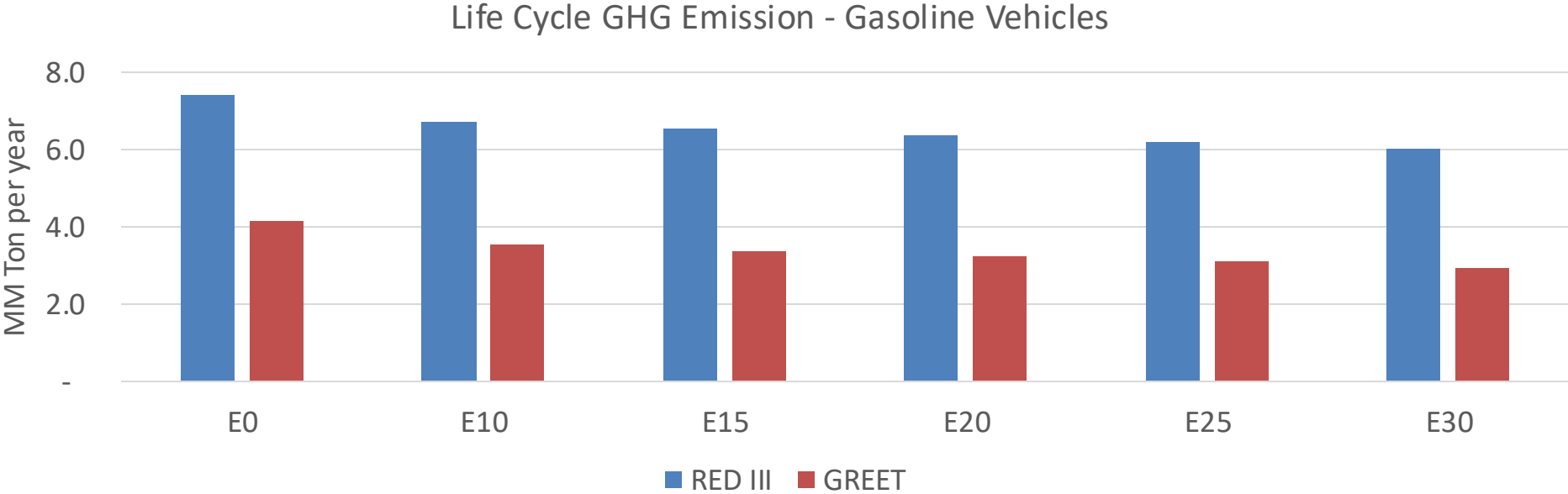
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Panama – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	23.8
2035 Target (MMT/year)	16.7
Delta	7.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	15.3%	19.2%	22.5%	25.8%	29.9%
RED II/III	0.0%	9.5%	12.0%	14.3%	16.4%	19.0%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.0%	11.2%	13.2%	15.1%	17.5%
RED II/III	0.0%	9.9%	12.5%	14.9%	17.1%	19.8%